Introduction to Mathematics and Optimisation

Exercises Week 12

Kirsten Pleskot (202507620)
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1 Problem 1

Let $f : \mathbb{R}^3 \to \mathbb{R}$ be the function given by

$$f(x, y, z) = x^2 + y^2 + (z - 1)^2.$$

Furthermore, let

$$\begin{aligned} g_1(x, y, z) &= z - x \\ g_2(x, y, z) &= z + x \\ g_3(x, y, z) &= z - y \\ g_4(x, y, z) &= z + y \\ g_5(x, y, z) &= -1 - z \end{aligned}$$

be functions from $\mathbb{R}^3$ to $\mathbb{R}$ and define

$$C = \{v \in \mathbb{R}^3 \mid g_1(v) \le 0, \quad g_2(v) \le 0, \quad g_3(v) \le 0, \quad g_4(v) \le 0, \quad g_5(v) \le 0\}$$

for $v = (x, y, z) \in \mathbb{R}^3$. In this problem, the focus is on the optimization problem

$$\begin{aligned} &\min f(v) \\ &v \in C. \end{aligned} \tag{1}$$

1. Explain why (1) is a convex optimization problem.
2. That $(x, y, z) \in C$ means that
$$z \le x, \quad z \le -x, \quad z \le y, \quad z \le -y, \quad z \ge -1. \tag{2}$$
Explain that from $z \le x$ and $z \le -x$ one can conclude that $z \le 0$. Likewise, explain why one can conclude that $-1 \le x \le 1$ and $-1 \le y \le 1$ based on the inequalities in (2).
3. Show that C is compact. Explain why this implies that the optimization problem (1) can be solved.

4. Is a solution to the optimization problem unique? If so, why?
5. Write down the KKT conditions corresponding to (1) in the variables $x, y, z, \lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5$ and explain from these why $(-1,-1,-1)$ cannot be an optimal solution to (1).
6. Show that $(0,0,0)$ is an optimal solution to (1). Do the KKT conditions corresponding to (1) have a unique solution?
7. Give a geometric interpretation of the optimization problem (1).

1.1 Convex optimization problem

According to Chapter 9 Convex optimization problem is one where we minimize function f over subset C where f is a convex function and C is a convex subset. To show that f is a convex function we can use Theorem 8.23 and show that the Hessian matrix of function f is positive definite.

The hessian matrix of f is:

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Which is positive definite by the result of Exercise 8.28 and therefore f is strictly convex by Theorem 8.23.

Since all g_i functions are linear by the result of Exercise 4.31 they are also convex. By Lemma 4.27 their preimages are convex subsets and by the result of Exercise 4.23 the intersection of those subsets is convex.

1.2 Rewriting inequities

By rewriting all of the inequalities as follows

$$\begin{aligned} z &\leq x \\ z &\leq -x \\ z &\leq y \\ z &\leq -y \\ z &\geq -1 \end{aligned}$$

And since $x \leq 0 \vee -x \leq 0$ then $z \leq 0$ by $z \leq x \wedge z \leq -x$.

And

$$\begin{aligned} z \leq x \wedge z \leq -x \wedge z \geq -1 &\Leftrightarrow \\ z \leq x \wedge x \leq -z \wedge -1 \leq z \wedge -z \leq 1 &\Leftrightarrow \\ -1 \leq z \leq x \leq -z \leq -1 &\Leftrightarrow \\ -1 \leq x \leq 1 & \end{aligned}$$

Similarly for y

1.3 Bounded set

Since $(-\infty, 0]$ is closed subset by Proposition 5.42 and set C is defined as intersection of preimages of closed subsets $\{(x, y, z) \in \mathbb{R}^3 \mid g_i(x, y, z) \in (-\infty, 0]\}$ then by Definition 5.39 and Definition 5.51 set C is closed.

By the previous part of the exercise $|x| \leq 1, |y| \leq 1, |z| \leq 1$ so by the Remark 5.30 the set is bounded

And since the set is bounded and closed it's compact by Definition 5.43

Therefore by Theorem 5.66 there exists a minimum for f under the set C

1.4 Uniqueness of solution

Since f is strictly convex it's minimum is global by Definition 6.13 and therefore unique

1.5 KKT Condition

The KKT conditions for function f are:

$$
\begin{aligned}
\lambda_1 &\geq 0 \\
\lambda_2 &\geq 0 \\
\lambda_3 &\geq 0 \\
\lambda_4 &\geq 0 \\
\lambda_5 &\geq 0 \\
\lambda_1(z - x) &= 0 \\
\lambda_2(z + x) &= 0 \\
\lambda_3(z - y) &= 0 \\
\lambda_4(z + y) &= 0 \\
\lambda_5(-1 - z) &= 0 \\
2x - \lambda_1 + \lambda_2 &= 0 \\
2y - \lambda_3 + \lambda_4 &= 0 \\
2z - 2 + \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 - \lambda_5 &= 0
\end{aligned}
$$

We can test point $(-1, -1, -1)$ and using the KKT condition show that it is not an optimal solution

$$\lambda_2(-1-1) = 0 \Leftrightarrow$$
$$\lambda_2 = 0 \Rightarrow$$
$$-2 - \lambda_1 + \lambda_2 = 0 \Leftrightarrow$$
$$-2 - \lambda_1 + 0 = 0 \Leftrightarrow$$
$$\lambda_1 = -2$$

This result breaks the condition $\lambda_2 \geq 0$ so by Theorem 9.34 (ii) the solution is not optimal

1.6 Optimal solution

At point $(0, 0, 0)$ the condition evaluate like so

$$\lambda_1 0 = 0$$
$$\lambda_2 0 = 0$$
$$\lambda_3 0 = 0$$
$$\lambda_4 0 = 0$$
$$\lambda_5(-1) = 0 \Leftrightarrow$$
$$\lambda_5 = 0$$

$$-\lambda_1 + \lambda_2 = 0$$
$$-\lambda_3 + \lambda_4 = 0$$
$$-2 + \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 - \lambda_5 = 0$$

$$-\lambda_1 + \lambda_2 = 0 \Leftrightarrow$$
$$\lambda_2 = \lambda_1$$

$$-\lambda_3 + \lambda_4 = 0 \Leftrightarrow$$
$$\lambda_3 = \lambda_4$$

$$\lambda_5 = 0 \wedge \lambda_3 = \lambda_4 \wedge \lambda_2 = \lambda_1 \Rightarrow$$
$$-2 + \lambda_1 + \lambda_1 + \lambda_3 + \lambda_3 = 0 \Leftrightarrow$$
$$\lambda_1 + \lambda_3 = 1$$

Which has infinetly many solution for $\lambda_1, \lambda_2, \lambda_3, \lambda_4,$

1.7 Geometrical interpretation

The function f describes a set of spheres with center at $(0, 0, -1)$ and the set C is a pyramid with the top at $(0, 0, 0)$ the problem is equivalent to finding the point in the pyramid closest to $(0, 0, -1)$